

Automated Water Requirement Estimator

Submitted in partial fulfillment of the requirements for the course

Microprocessors and Microcontrollers (BECE204L)

by

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1. INTRODUCTION

1.1. Objective

The primary objective of this project is to develop an **automated water management system** for paddy fields using an **ESP32 microcontroller** and multiple sensors to optimize irrigation based on real-time environmental data. The system aims to:

1. **Accurately measure key environmental factors** such as **rainfall, soil moisture, and temperature** using respective sensors.
2. **Implement a regression-based water requirement model** to determine the optimal amount of water needed for a 1 acre paddy field.
3. **Display real-time water requirement calculations** on an **OLED screen**, providing instant feedback to farmers.
4. **Reduce water wastage** by ensuring that irrigation is provided only when necessary, aligning with sustainable farming practices.
5. **Enhance efficiency in agricultural water management** by leveraging IoT technology to automate and simplify decision-making.

This system contributes to **sustainable agricultural practices** by conserving water resources while ensuring optimal crop growth, aligning with **UN Sustainable Development Goals (SDG 6 & 12)** for water conservation and responsible resource use.

1.2. Problem Statement

Agriculture is one of the largest consumers of freshwater resources, with paddy fields requiring significant amounts of water for optimal crop growth. Traditional irrigation methods often lead to excessive water usage, inefficient distribution, and wastage, negatively impacting both **water conservation** and **crop yield**. In regions with unpredictable rainfall, farmers struggle to determine the precise amount of water required, leading to either over-irrigation or water scarcity.

To address this issue, this project develops an **automated water management system** using an **ESP32 microcontroller** and **sensor-based data collection**. By integrating real-time measurements of **rainfall, soil moisture, and temperature**, the system **calculates the optimal water requirement** for a **1 acre paddy field** using a **regression-based approach**. This ensures that irrigation is applied efficiently, reducing water wastage while maintaining ideal conditions for crop growth.

Relevance to UN Sustainable Development Goals (SDGs)

1. SDG 6: Clean Water and Sanitation

- Promotes sustainable water management by **optimizing irrigation practices** and reducing **water wastage in agriculture**.

2. SDG 12: Responsible Consumption and Production

- Encourages efficient use of natural resources through **smart irrigation systems**, preventing unnecessary depletion of freshwater reserves.

3. SDG 13: Climate Action

Helps in adapting to **climate variability** by using real-time environmental data to **adjust irrigation patterns**, making agriculture more resilient.

By integrating **IoT-based smart farming techniques**, this project contributes to **sustainable water resource management**, improving agricultural efficiency while minimizing environmental impact

1.3. Key contributions

This project presents a **smart irrigation system** that leverages IoT and **sensor-based automation** to optimize water usage in paddy fields. The key contributions of this work include:

1. Development of a Regression-Based Water Management Model

- Designed a **multiple regression equation** that calculates the **optimal water requirement** based on real-time **rainfall, soil moisture, and temperature data**.

2. Implementation of an IoT-Based Automated System

- Integrated **ESP32 microcontroller** with **ultrasonic, soil moisture, and temperature sensors** to collect real-time environmental data for precise irrigation decisions.

3. Real-Time Water Requirement Calculation and Display

- Displayed the computed **water requirement in liters** for a **1000 m² paddy field** on an **OLED screen**, allowing farmers to take immediate action.

4. Reduction of Water Wastage and Improved Efficiency

- Enabled **smart decision-making** for irrigation, ensuring water is supplied only when necessary, contributing to **water conservation and sustainable agriculture**.

This project demonstrates how **IoT and data-driven decision-making** can enhance **water efficiency in agriculture**, making farming practices more **sustainable, cost-effective, and resource-efficient**.

2. SYSTEM MODEL

2.1. Proposed Methodology

The proposed system utilizes IoT-based smart irrigation techniques to optimize water usage in paddy fields. The methodology consists of the following steps:

1. Sensor Data Collection

- Rainfall Measurement: An ultrasonic sensor measures the water level in a calibrated rain collection container and converts it into mm of rainfall.
- Soil Moisture Detection: A soil moisture sensor records moisture levels in the soil, indicating how much additional water is needed.
- Temperature Monitoring: A DHT22 sensor measures temperature, which influences evaporation rates and water requirement calculations.

2. Regression-Based Water Requirement Calculation

- A multiple regression equation is used to estimate the optimal water requirement based on the collected sensor data.
- The equation considers rainfall (R), soil moisture (S), and temperature (T) to calculate the necessary irrigation volume (W) for a 1 acre paddy field.

3. Processing and Display

- The ESP32 microcontroller processes sensor data and computes the required water supply.
- The result is displayed on an OLED screen, providing real-time guidance for farmers.

4. Real-Time Monitoring and Decision Support

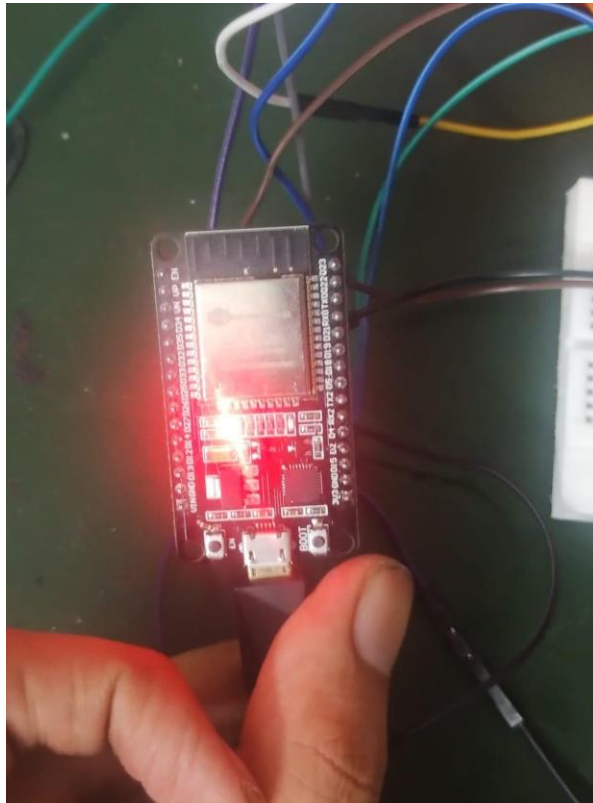
- The system continuously updates the calculations based on real-time environmental conditions.
- Farmers can use this information to decide when and how much water to supply, reducing wastage and ensuring optimal crop hydration.

5. Scalability and Future Enhancements

- The system can be expanded by integrating wireless connectivity for remote monitoring and automation.
- Future enhancements may include AI-based predictions for irrigation scheduling and mobile app integration for remote access.

This methodology ensures a data-driven approach to agricultural water management, helping to conserve water, enhance efficiency, and promote sustainable farming practices.

2.2 Analytical and Theoretical Description



1) ESP 32 :

The **ESP32** is a powerful **microcontroller** with **Wi-Fi and Bluetooth** capabilities, making it ideal for **IoT applications**. In this project, the ESP32 acts as the **central processing unit**, collecting data from the **ultrasonic sensor (rainfall)**, **soil moisture sensor**, and **DHT22 (temperature)**. It processes the sensor readings, applies the **regression model** to compute the required **water supply**, and displays the result on an **OLED screen**. Additionally, its **low power consumption**, multiple **GPIO pins**, and **real-time data handling** make it an efficient choice for **smart irrigation systems**.

2) Temperature Sensor



The **DHT22 temperature sensor** is used in this project to measure **ambient temperature**, which plays a crucial role in determining the **water requirements** for paddy fields. It provides **high accuracy** temperature readings ranging from **-40°C to 80°C** with an accuracy of **±0.5°C**. The sensor communicates with the **ESP32** via a **single digital pin**, ensuring easy integration. Since temperature affects **evaporation rates**, its data is used in the **regression model** to calculate the optimal amount of water needed for irrigation.

3) Soil Moisture:



The **soil moisture sensor** is used to measure the **water content in the soil**, which helps determine whether additional irrigation is required. It consists of two probes that detect **soil conductivity**, where **higher moisture levels** result in **lower resistance** and vice versa. In this project, the sensor provides **analog output (A0)** to the **ESP32**, which processes the data and incorporates it into the **regression model** to calculate the optimal **water supply**. By continuously monitoring soil moisture, the system helps in **efficient water management** and prevents **overwatering or underwatering** of the paddy field.

4) Method of Measuring Rainfall using Using Ultrasonic sensor



In this project, an **ultrasonic sensor (HC-SR04)** is used to measure the **rainfall received** by detecting the water level in a **rain collection can**. The process involves the following steps:

Rainwater Collection: A **cylindrical container** (13 cm diameter, 14 cm height) is placed in the open field to collect rainfall.

Ultrasonic Measurement: The **HC-SR04 sensor** is mounted at the **top of the container**, facing downward. It emits **ultrasonic waves** and measures the **time taken** for the echo to bounce back from the water surface.

Water Level Calculation: Using the **speed of sound (343 m/s)**, the **distance to the water surface** is determined. The depth of the water is then calculated as:

$$\text{Water Height} = \text{Container Height} - \text{Measured Distance}$$

This method provides an **accurate and cost-effective** way to measure rainfall using a **microcontroller-based system**.

5) Regression Line Explanation

The project's regression equation estimates the **water required (W)** for a **1 acre paddy field** based on:

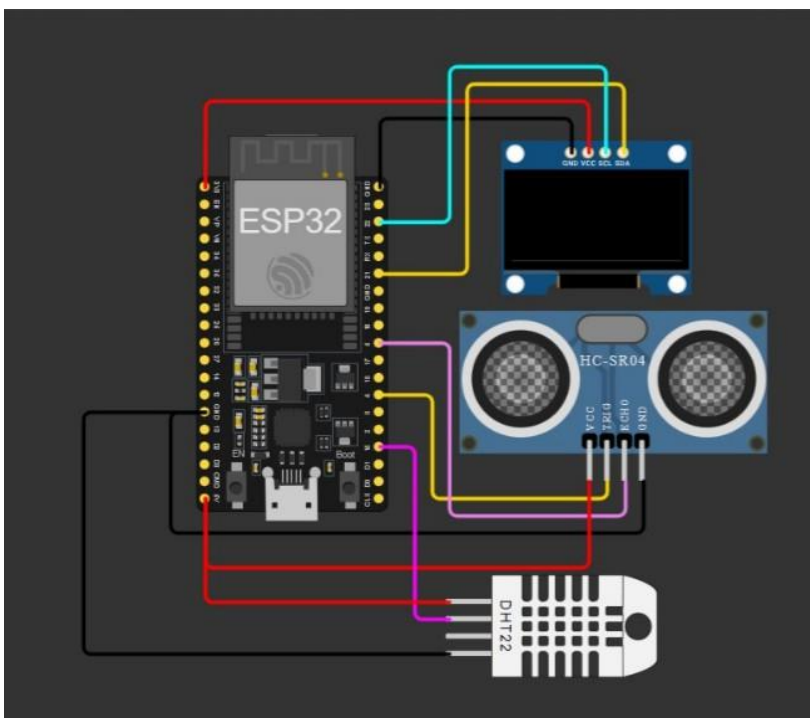
$$W=70-0.9R-0.7S+1.5T$$

where:

- **R** = Rainfall (mm) (↓ less water needed)
- **S** = Soil moisture (%) (↓ less water needed)
- **T** = Temperature (°C) (↑ more water needed)

The **ESP32** collects real-time sensor data, substitutes it into the equation, and calculates **W**, which is displayed on an **OLED screen**. This ensures **efficient water management** using real-time environmental conditions.

6) Simulated Block Diagram

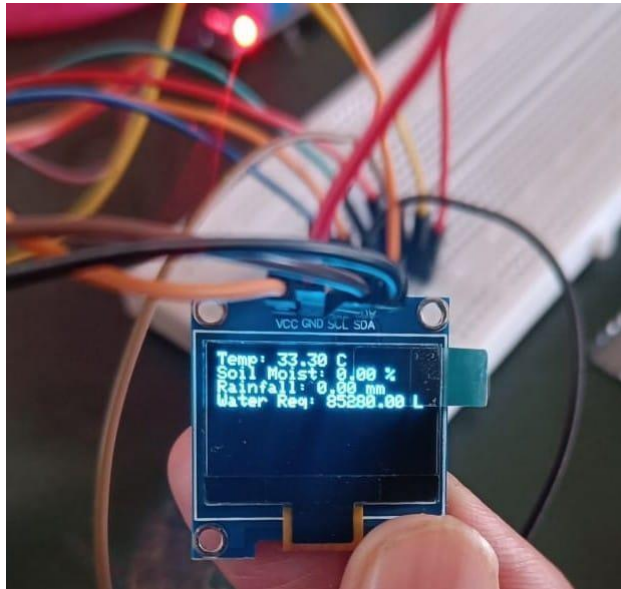


3. RESULTS & DISCUSSIONS

3.1. Results & Discussions

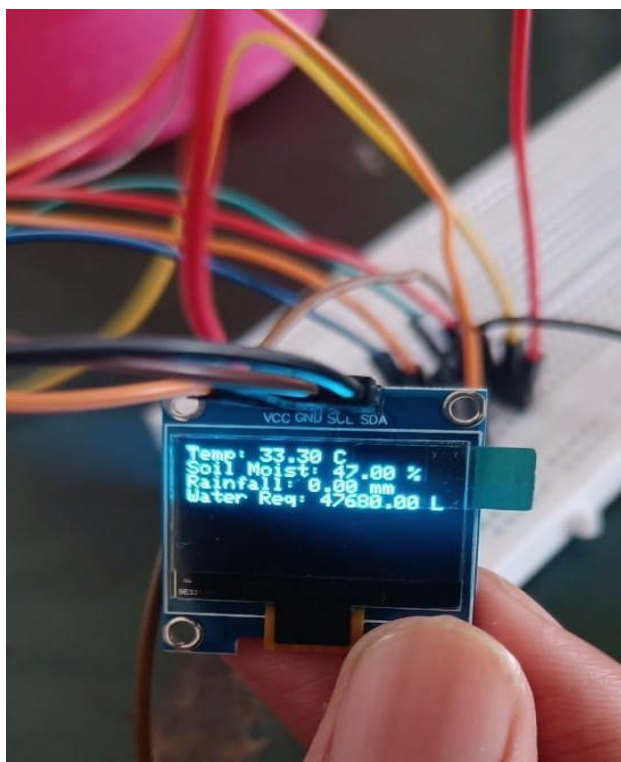
1) Case 1 :

When no rainfall is received and the soil has no moisture content then 1 acre of paddy field approximately requires 85,000 litres of water.



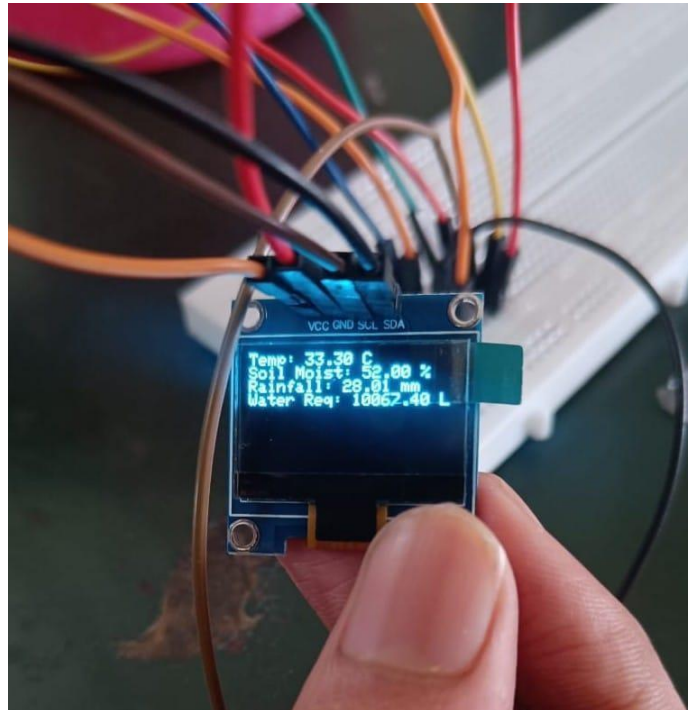
2) Case 2 :

When there is some soil moisture even though it hasn't received any rainfall then the amount of water required reduces to some extent depending upon the amount of soil moisture.



3) Case 3 :

When the paddy field has received rainfall and the soil moisture is also increased to some extent then the paddy field considerably requires less water to be supplied on that day.



Final Circuit Diagram :



4. CONCLUSION

This project presents an automated water management system for paddy fields using ESP32 and multiple sensors. By integrating real-time rainfall, soil moisture, and temperature data, the system calculates the precise water requirement using a regression model. The collected data is processed and displayed on an OLED screen, allowing farmers to make informed irrigation decisions. This approach optimizes water usage, prevents over-irrigation, and enhances sustainable agriculture.

Future Work

- **Machine Learning Integration:** Improve the accuracy of water requirement predictions by training the model with real-world agricultural datasets.
- **Additional Sensors:** Incorporate humidity, wind speed, and evaporation rate sensors for a more comprehensive irrigation strategy.
- **IoT and Remote Monitoring:** Develop a cloud-based dashboard or mobile app to allow farmers to remotely monitor and control irrigation.
- **Automated Irrigation:** Link the system to an electric valve or motor pump to automate water supply based on calculated needs.
- **Scalability:** Adapt the system for various crops and larger agricultural fields, making it useful for diverse farming environments.

This project aligns with UN Sustainable Development Goals (SDG) 6 (Clean Water and Sanitation) and SDG 13 (Climate Action) by promoting efficient water management and climate-resilient agriculture. Through further enhancements, this system can significantly contribute to sustainable farming practices, improving water conservation and food security.

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